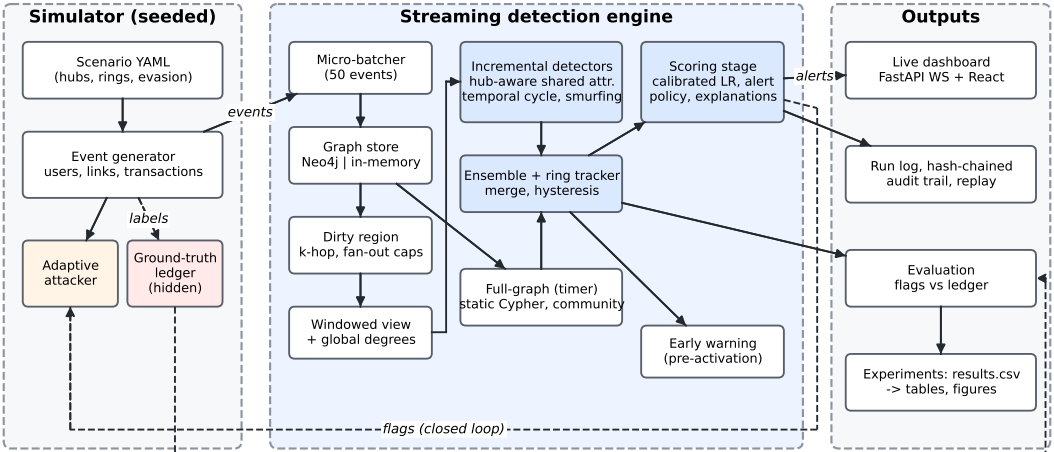
Simulator (seeded)
Scenario YAML
(hubs, rings, evasion)
Event generator
users, links, transactions
Adaptive
attacker
Ground-truth
ledger
(hidden)
events
labels
Streaming detection engine
Micro-batcher
(50 events)
Graph store
Neo4j | in-memory
Dirty region
k-hop, fan-out caps
Windowed view
+ global degrees
Incremental detectors
hub-aware shared attr.
temporal cycle, smurfing
Ensemble + ring tracker
merge, hysteresis
Full-graph (timer)
static Cypher, community
Scoring stage
calibrated LR, alert
policy, explanations
Early warning
(pre-activation)
alerts
Outputs
Live dashboard
FastAPI WS + React
Run log, hash-chained
audit trail, replay
Evaluation
flags vs ledger
Experiments: results.csv
-> tables, figures
flags (closed loop)